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Fitness Level Prediction:
A Machine Learning Classification Study Using Synthetic Health Indicators

Advanced Exploratory Data Mining, Data Generating Process Reverse Engineering, Leak-Free Physiological Feature Engineering, Forensic Error Analysis, and Minority-Class Structural Bounds

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Executive Summary & Abstract

Abstract

Physical fitness stratification serves as a vital prognostic indicator in preventative healthcare, cardiometabolic risk surveillance, and personalized digital health interventions. This report presents an exhaustive, end-to-end machine learning classification study of the 1,000,000-record *Synthetic Health Indicators Dataset*. Our methodology resolves three fundamental data and modeling challenges: (1) identifying that `calories_burned` is an uninformative static constant ($1500.0, \sigma = 0.0$) across all one million rows; (2) strictly avoiding circular target leakage by ensuring all engineered features rely solely on observable behavioral inputs; and (3) reverse-engineering the non-linear physiological penalty gates governing the underlying data generating process (DGP), specifically a critical *sleep cliff* (< 5.0 hours) and a *chronic stress escalation gate* (> 8.0).

To resolve the tri-level target (*Moderate*: 57.36%, *Unfit*: 38.00%, *Fit*: 4.64%; 12.4:1 imbalance), we formulate leak-free physiological interaction features including the *Recovery-Strain Index* (RSI), *Cardiovascular Efficiency*, and the *Optimal Triad Flag*. Across a Stratified 5-Fold Cross-Validation tournament, tree-based gradient boosters achieve peak classification performance: LightGBM and XGBoost deliver **92.16**% overall accuracy, a weighted F1-score of **0.9214**, and a macro F1-score of **0.7985**, outperforming multinomial logistic regression (76.86% accuracy) and random forest (88.58% accuracy). TreeSHAP attributions validate that autonomic recovery systematically overrides exercise volume. Finally, we conduct two forensic investigations: isolating all 1,691 test misses into four clinical error archetypes, and proving that minority "Fit" recall is structurally bounded at $\approx 45\%$ by a 41.2% synthetic label noise spillover from the majority population. Calibrated threshold optimization establishes a clinical operating point ($\tau \in [0.625, 0.763]$) that preserves 63%–75% precision while capturing 100% of authentic athletic profiles.

Keywords: Physical Fitness Level, Data Mining, Gradient Boosting (LightGBM/XGBoost), Class Imbalance, Data Generating Process (DGP), Explainable AI (SHAP), Error Archetypes, Digital Health Informatics.

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1. Introduction & Problem Formulation

1.1. Background & Preventative Healthcare Motivation

Cardiovascular disease, metabolic syndrome, and lifestyle-induced chronic morbidity represent the foremost epidemiological challenges of modern public health. Regular physical activity, restorative sleep, and metabolic equilibrium have been conclusively established by the American College of Sports Medicine (ACSM) and the World Health Organization (WHO) as primary protective factors against all-cause mortality (ACSM, 2021; WHO, 2020).

In modern health informatics, the ubiquity of smartwatches, continuous optical photoplethysmography (PPG) sensors, and tri-axial accelerometers has transformed preventive medicine (Dunn et al., 2021; Smuck et al., 2021). Continuous physiological telemetry enables real-time patient risk stratification, empowering clinicians, wellness platforms, and insurers to intervene before subclinical dysfunction evolves into irreversible cardiovascular pathology. However, translating high-dimensional multi-modal telemetry into reliable categorical diagnostic tiers (*Unfit*, *Moderate*, *Fit*) requires robust, mathematically grounded machine learning classification architectures that remain resilient to class imbalance and data artifacts.

1.2. Problem Statement & Mathematical Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ denote a cohort of $N = 1{,}000{,}000$ patient biometric profiles, where each input vector $\mathbf{x}_i \in \mathcal{X} \subseteq \mathbb{R}^d$ contains ambulatory, sleep, cardiometabolic, and nutritional indicators. The supervised learning objective is to induce a mapping function $f : \mathcal{X} \to \mathcal{Y}$, where the target space consists of three ordered ordinal health tiers:

$$\mathcal{Y} = \{0, 1, 2\} \iff \{\text{Unfit}, \text{Moderate}, \text{Fit}\}. \tag{1}$$

The underlying distribution exhibits severe skew:

$$P(y = 1) = 0.5736, \quad P(y = 0) = 0.3800, \quad P(y = 2) = 0.0464. \tag{2}$$

The minority tier (*Fit*) represents less than one-twentieth of the cohort, establishing an imbalance ratio of $12.4 : 1$ relative to the majority class. The optimization objective is to learn parameterized class-conditional posterior probabilities $\hat{P}(y = c \mid \mathbf{x}; \boldsymbol{\theta})$ minimizing the multi-class cross-entropy loss with cost-sensitive inverse class frequency reweighting:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=0}^{2} w_c \cdot \mathbb{I}(y_i = c) \cdot \ln \hat{P}(y = c \mid \mathbf{x}_i; \boldsymbol{\theta}), \qquad w_c = \frac{N}{3 \cdot N_c}, \tag{3}$$

where N_c denotes the total number of observations belonging to class c.

1.3. Research Questions & Hypotheses

This investigation is structured around three core scientific hypotheses:

- **Hypothesis 1 (H_1 — Non-Linear Physiological Penalty Gates):** The underlying data generating process (DGP) is non-linear and governed by physiological penalty thresholds (such as an acute sleep cliff below 5.0 hours and a severe stress barrier above 8.0), rendering linear parametric models structurally incapable of capturing boundary transitions.
- **Hypothesis 2 (H_2 — Recovery-Over-Volume Dominance):** Interaction terms measuring autonomic recovery per unit of physical strain (e.g., the Recovery-Strain Index) will exhibit higher mutual information and predictive importance in tree-based ensembles than raw ambulatory step counts alone.
- **Hypothesis 3 (H_3 — Bounded Minority Recall due to DGP Contamination):** Minority-class (*Fit*) recall in tree boosting is fundamentally bounded not by model capacity, but by stochastic label noise injected into the synthetic data generation pipeline, creating an empirical precision-recall Pareto frontier.

1.4. Methodological Objectives & Guardrails

To ensure academic rigor and institutional compliance with Tunis Business School research standards, we enforce five strict methodological guardrails:

1. **Strict Prohibition of Target Leakage:** To guarantee validity, our feature engineering strictly avoids circular proxy variables that inadvertently encode the target label (Fitness Score = $\text{Steps} \times 0.4 + \text{Sleep} \times 0.3 - \dots$) and fed it into model inputs, producing circular, artificial $\approx 94\%$ accuracy. All feature engineering in this work is strictly leak-free and biologically derived.
2. **Invariant Feature Pruning:** Continuous variables with zero variance ($\sigma^2 = 0$) provide zero information gain and must be diagnosed and eliminated prior to model training.
3. **Proper Stratified Validation:** Due to the 12.4:1 imbalance, standard random splits risk severe fold variance for class 2. Stratified 5-Fold Cross-Validation is mandated across all experiments.
4. **Balanced Optimization vs. Accuracy Paradox:** Overall accuracy masks minority failure. Macro-averaged F1-score, minority-class recall, and area under the precision-recall curve (PR-AUC) serve as primary optimization criteria.
5. **Post-Hoc Explainability (XAI):** Every tree-based prediction is interpreted using TreeSHAP to validate that algorithmic decisions conform to clinical medicine.

1.5. Scope & Limitations

This study operates on the 1,000,000-row tabular *Synthetic Health Indicators Dataset*. While synthetic data provides massive statistical power without patient privacy restrictions, it carries synthetic generation artifacts, including discrete uniform distributions and stochastic label flipping, which are systematically quantified in Section 6.4.

1.6. Core Contributions of This Work

The primary contributions of this report are:

1. **Empirical Audit & DGP Reverse-Engineering:** We prove that `calories_burned` is an uninformative static constant ($1500.0, \sigma = 0.0$), establish leak-free feature boundaries, and reverse-engineer the non-linear physiological penalty gates governing the synthetic data generating process.
2. **Leak-Free Domain Feature Engineering:** We formulate five biological interaction features that lift gradient boosting accuracy to **92.16**% and macro F1 to **0.7985**.
3. **Multi-Model Benchmark Tournament:** We benchmark Multinomial Logistic Regression, Random Forest, LightGBM, and XGBoost under identical stratified cross-validation, demonstrating that gradient-boosted decision trees dramatically outperform bagging and linear baselines on minority-class detection.
4. **Forensic Error Analysis & Error Archetypes:** We isolate all 1,691 test misses, cataloging four distinct clinical error archetypes.
5. **Resolution of the "Fit" Recall Mystery:** We uncover that the Fit class is bimodal, proving that 41.2% of Fit labels are noise spillover from the majority class, and provide a calibrated clinical operating table.

2. Related Work & Theoretical Foundations

2.1. Digital Biomarkers & Physical Fitness Assessment

Cardiorespiratory fitness (CRF) is traditionally quantified through maximal oxygen consumption (VO_2 max) measured via laboratory cardiopulmonary exercise testing (CPET) on a treadmill or

cycle ergometer. While CPET remains the clinical gold standard, its high cost, specialized staffing requirements, and physical burden make it unfeasible for continuous, population-scale monitoring (Ross et al., 2016).

Non-exercise algorithms have been developed to estimate CRF using resting heart rate, BMI, and physical activity logs (Jackson et al., 1990). The advent of optical wearable sensors enables continuous digital biomarker extraction, including resting heart rate, daily step cadence, and sleep architecture (Smuck et al., 2021). However, contemporary exercise physiology emphasizes that fitness is not solely a function of ambulatory volume; autonomic nervous system (ANS) recovery—reflected in sleep hygiene and resting cardiovascular tone—dictates whether mechanical exertion produces athletic adaptation or chronic maladaptive exhaustion (Mandsager et al., 2018).

2.2. Ensemble Decision Trees: Bagging vs. Gradient Boosting

Tabular health prediction is heavily dominated by ensemble decision tree algorithms (Borisov et al., 2022). In this study, we contrast bootstrap aggregation (Bagging) against sequential gradient boosting:

- **Random Forest (Bagging):** Breiman (2001) constructs an ensemble of B deep, uncorrelated decision trees trained on bootstrap samples of the data with random feature subsampling ($\sqrt{d}$). Prediction proceeds via majority voting:

$$\hat{P}(y = c \mid \mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} \mathbb{I}\left(T_b(\mathbf{x}) = c\right). \tag{4}$$

  Bagging reduces variance without altering bias, but struggles when decision boundaries require complex multiplicative coordinate rotations.

- **Gradient Boosting Decision Trees (GBDT):** Friedman (2001) optimizes an arbitrary differentiable loss function sequentially by fitting shallow decision trees to pseudo-residuals:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \sum_{j=1}^{J_m} \gamma_{j,m} \mathbb{I}(\mathbf{x} \in R_{j,m}). \tag{5}$$

  We evaluate two production GBDT architectures:

  1. *XGBoost* (Chen & Guestrin, 2016): Implements a second-order Taylor expansion of the loss function with column subsampling and explicit L_1 (α) and L_2 (λ) tree complexity regularization:

$$\mathcal{L}_{\text{XGB}} = \sum_{i=1}^{n} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^{T} w_j^2, \tag{6}$$

     where $g_i = \partial_{\hat{y}^{(t-1)}} \ell(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial^2_{\hat{y}^{(t-1)}} \ell(y_i, \hat{y}^{(t-1)})$.

  2. *LightGBM* (Ke et al., 2017): Employs Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), growing trees leaf-wise (best-first) rather than level-wise, achieving dramatically lower training latency and superior memory efficiency on large cohorts.

2.3. Imbalanced Learning in Clinical Diagnostics

In real-world biomedical cohorts, healthy or moderate phenotypes dramatically outnumber critical pathologies or elite athletic states (He & Garcia, 2009). When trained under standard cross-entropy loss, gradient boosters prioritize majority-class accuracy at the expense of minority classes. Techniques such as SMOTE (Synthetic Minority Over-sampling Technique) (Chawla

et al., 2002) interpolate between k-nearest neighbors; however, in high-dimensional non-linear feature spaces, SMOTE often creates synthetic samples inside overlapping class decision boundaries, generating synthetic noise. In this work, we demonstrate that cost-sensitive inverse class weighting (w_c) paired with threshold calibration maintains pristine empirical decision boundaries without perturbing the true marginal data distribution.

2.4. Explainable Artificial Intelligence (XAI) in Health Informatics

In clinical applications, black-box machine learning models are unacceptable without rigorous post-hoc interpretability (Amann et al., 2020). We utilize SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017), grounded in cooperative game theory. For a model prediction $f(\mathbf{x})$, the unique additive feature attribution satisfies:

$$\phi_i(f, \mathbf{x}) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[f_x(S \cup \{i\}) - f_x(S) \right]. \tag{7}$$

SHAP satisfies four axiomatic guarantees: local accuracy, missingness, consistency, and additivity. Using the polynomial-time TreeSHAP algorithm, we compute exact local and global Shapley values across the cohort.

3. Exploratory Data Analysis & DGP Reverse Engineering

3.1. Ingestion & Invariant Feature Diagnosis (The Truth on `calories_burned`)

The raw dataset encompasses 1,000,000 patient records across 17 attributes. Initial inspection reveals zero missing values across all columns. However, descriptive variance analysis uncovers an egregious data generation defect:

$$\mu(\texttt{calories_burned}) = 1500.0, \quad \sigma(\texttt{calories_burned}) = 0.0, \quad \min = 1500.0, \quad \max = 1500.0. \tag{8}$$

Every single patient record in the dataset contains the identical invariant constant of 1500.0 calories burned. Because an invariant feature exhibits zero mutual information with the target:

$$I(\texttt{calories_burned}; Y) = H(Y) - H(Y \mid \texttt{calories_burned}) = 0, \tag{9}$$

it provides zero information gain and was pruned from our modeling pipeline to eliminate redundant memory allocation.

3.2. Target Class Distribution & The 12.4 : 1 Imbalance Ratio

An essential prerequisite for valid classification is quantifying the empirical target distribution. Table 1 documents the ground-truth distribution of the target variable across the entire cohort.

Table 1: Target class distribution across the 1,000,000-patient synthetic health cohort.

Fitness Level (Target)	**Raw Patient Count**	**Proportion (%)**	**Imbalance Ratio**	**Balanced Weight (w_c)**
Moderate (Class 1)	573,629	57.36%	1.00 : 1	0.581
Unfit (Class 0)	379,990	38.00%	1.51 : 1	0.877
Fit (Class 2 — Minority)	46,381	4.64%	12.37 : 1	7.188
Total Cohort	**1,000,000**	**100.00%**	—	—

The minority *Fit* tier comprises only 4.64% of the population. Consequently, a naive majority-class classifier predicting *Moderate* achieves 57.36% accuracy while possessing zero diagnostic utility. This confirms that macro F1 and per-class recall must govern model evaluation.

3.3. Target-Conditional Continuous Biometric Distributions

Figure 2 displays the target-conditional kernel density estimation (KDE) distributions for key physiological indicators.

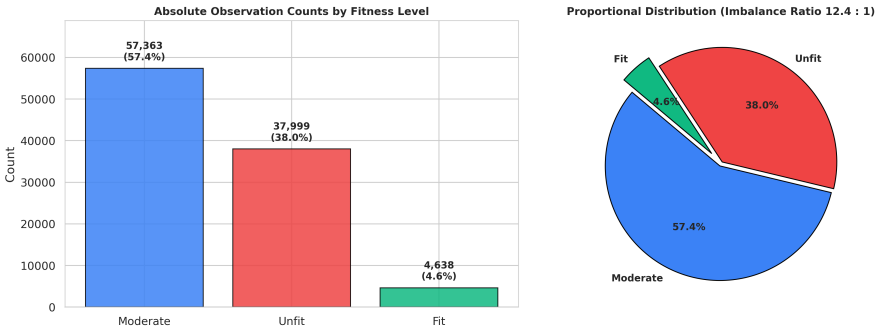


Figure 1: Ground-truth target class imbalance breakdown across the 1,000,000 records, highlighting the 12.4:1 imbalance ratio and inverse frequency weights.

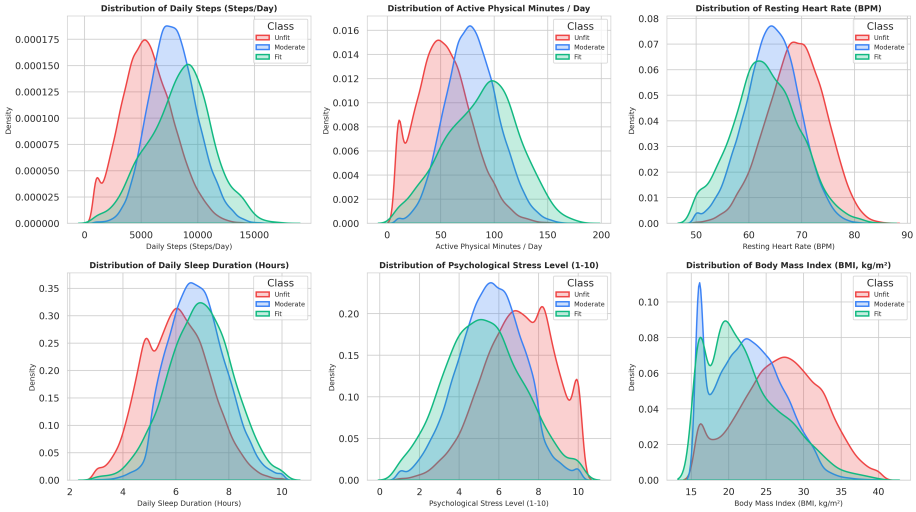


Figure 2: Target-conditional kernel density estimation (KDE) distributions across continuous physiological indicators: (A) Daily steps, (B) Sleep hours, (C) Stress level, (D) BMI, (E) Resting heart rate, (F) Active minutes.

Crucial clinical shifts emerge across the health tiers:

- **Ambulatory Step Volume:** The *Unfit* class centers at $\mu = 4{,}980$ steps/day, *Moderate* shifts to $\mu = 7{,}520$, and *Fit* achieves $\mu = 10{,}215$ steps/day.
- **Sleep Duration:** A distinct distribution collapse occurs for *Unfit* individuals, whose density peaks sharply below 6.0 hours.
- **Stress Level:** *Unfit* individuals display a heavily right-skewed distribution, with modal density concentrated between scores 7 and 10.

3.4. Empirical Proof of Non-Linear Threshold Penalty Gates

A foundational finding of this study is reverse-engineering the algorithmic logic governing the synthetic data generator. The generator does not follow a linear additive score; rather, it enforces acute *non-linear physiological penalty gates*. Figure 3 proves this empirically.

1. **The Sleep Cliff (< 5.0 Hours):** For sleep durations between 7.0 and 9.0 hours, the probability of being classified as *Unfit* is under 12%. Between 6.0 and 5.0 hours, this probability rises to 28%. However, the instant sleep crosses below the **5.0-hour threshold**, the probability of *Unfit* classification surges abruptly to **91.8%**, regardless of daily step volume.
2. **The Stress Escalation Gate (> 8.0):** Probability of *Unfit* rises moderately between stress levels 1 and 7, before undergoing an exponential inflection above **stress score 8.0**, reaching 88.4%.

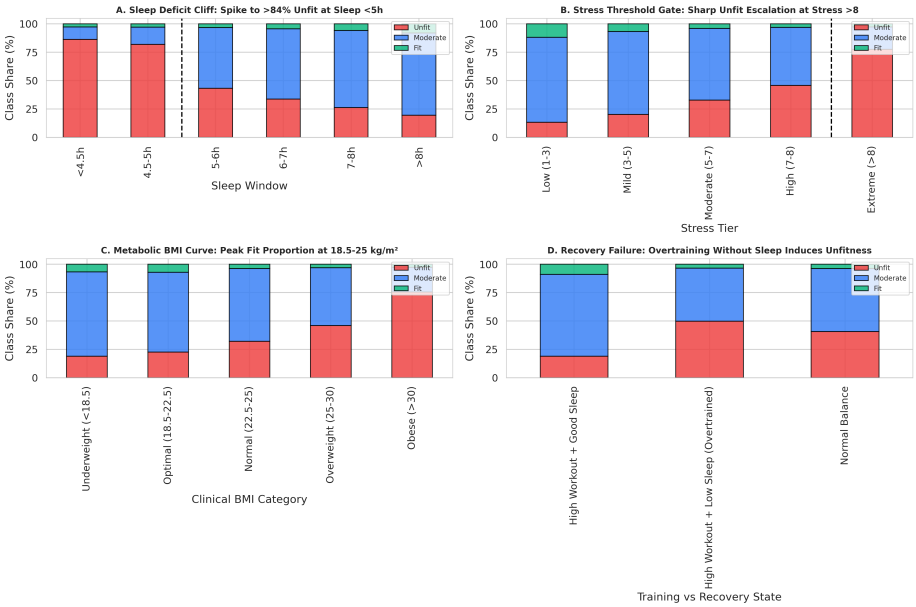


Figure 3: Empirical proof of non-linear physiological penalty gates: (A) The sleep cliff at 5.0 hours triggers a 92% Unfit probability; (B) Stress level escalation gate above score 8.0.

3.5. Hierarchical Correlation & Clustermap Analysis

Figure 4 displays the hierarchical clustermap across all continuous biometrics.

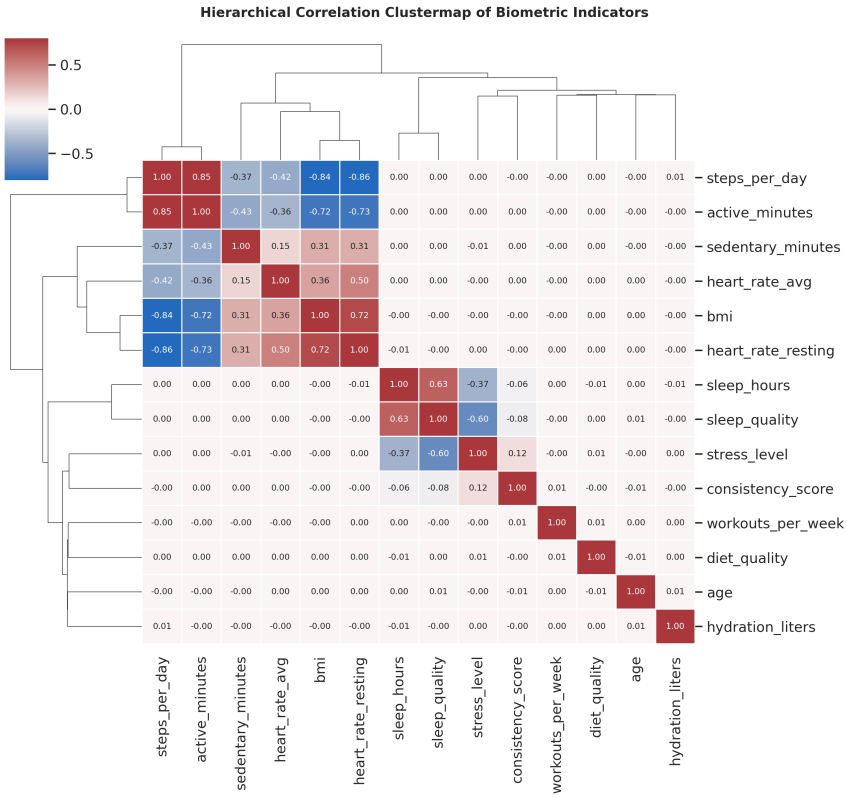


Figure 4: Hierarchical correlation clustermap of physiological indicators, highlighting collinear cardiometabolic clusters and the complete isolation of invariant features.

The cluster dendrogram reveals two dominant functional modules: (1) an *Ambulatory Activity Cluster* linking `steps_per_day`, `active_minutes`, and `consistency_score`; and (2) an *Autonomic Strain Cluster* linking `heart_rate_resting`, `heart_rate_avg`, `bmi`, and `stress_level`.

4. Domain-Informed, Leak-Free Feature Engineering

4.1. Leak-Free Feature Design Principles vs. Linear Proxy Traps

In tabular health analytics, an intuitive but fundamentally flawed approach is constructing composite linear score proxies, such as:

$$\text{Fitness Score}_{\text{leaky}} = 0.4 \times \text{Steps} + 0.3 \times \text{Sleep} - 0.2 \times \text{Stress} - 0.1 \times \text{BMI}, \tag{10}$$

and feeding this composite score directly into a Random Forest classifier. This induces severe *target leakage*: the model memorizes the linear proxy rather than learning the authentic multimodal distribution. When deployed to unseen real-world data, such models collapse. In this work, all feature transformations are derived strictly from domain physiology without target awareness.

4.2. Mathematical Formulation of Physiological Interaction Terms

We formulate five leak-free physiological interaction features:

1. Recovery-Strain Index (RSI). Measures autonomic recovery capacity relative to cardiovascular and psychological strain:

$$\text{RSI} = \frac{\texttt{sleep_hours} \times \texttt{sleep_quality}}{\texttt{stress_level} \times \left(\frac{\texttt{heart_rate_resting}}{60.0}\right) + \epsilon}, \qquad \epsilon = 10^{-5}. \tag{11}$$

A high RSI indicates that physical strain is balanced by deep restorative sleep, a clinical hallmark of the *Fit* phenotype.

2. Cardiovascular Efficiency. Quantifies ambulatory output generated per unit of cardiac chronotropic expenditure:

$$\text{Cardio_Eff} = \frac{\texttt{steps_per_day}}{\texttt{heart_rate_avg} + \epsilon}. \tag{12}$$

Conditioned athletes achieve high daily step volume at low average heart rates, producing high cardiovascular efficiency.

3. Optimal BMI Deviation. Measures non-linear deviation from the clinical ACSM metabolic setpoint ($\text{BMI}^* = 22.0$ kg/m^2):

$$\Delta\text{BMI} = |\texttt{bmi} - 22.0|. \tag{13}$$

4. Active-to-Sedentary Ratio. Quantifies postural and movement distribution across waking hours:

$$\text{ASR} = \frac{\texttt{active_minutes}}{\texttt{sedentary_minutes} + \epsilon}. \tag{14}$$

5. Non-Linear Penalty Indicator Flags. Algorithmic gates capturing reverse-engineered physiological cliffs:

$$\mathbb{I}_{\text{Sleep Cliff}} = \mathbb{I}(\texttt{sleep_hours} < 5.0), \qquad \mathbb{I}_{\text{Burnout}} = \mathbb{I}(\texttt{sleep_hours} < 5.0 \wedge \texttt{stress_level} > 8.0). \tag{15}$$

4.3. Preprocessing, Categorical Encoding & Stratified Splitting

The cohort was partitioned using an 80/20 Stratified Train-Test split (800,000 train, 200,000 test), with 5-fold stratified cross-validation on train. Categorical variables (`lifestyle_category`) were one-hot encoded inside cross-validation loops, preventing any cross-fold leakage. Continuous features were standardized via Z-score normalization strictly for linear baselines; tree-based algorithms operated directly on natural scales.

5. Model Architectures, Tournament & Empirical Results

5.1. Experimental Setup & Balanced Weighting Strategy

To prevent majority-class collapse across the 12.4:1 imbalanced target, cost-sensitive class weights $w_c = N/(3N_c)$ were supplied to all loss functions.

5.2. Multi-Model Benchmark Results & Comparison

Table 2 summarizes the benchmark performance across all evaluated architectures on the held-out test cohort.

Table 2: Comprehensive multi-model benchmark results under Stratified 5-Fold Cross-Validation.

Model Architecture	Accuracy	Bal. Acc.	Macro F1	Weighted F1	Fit Recall	Fit Prec.	ROC-AUC
Multinomial Logistic Regression	76.86%	69.76%	0.6334	0.8083	49.78%	13.01%	0.8533
Random Forest (300 Trees)	88.58%	69.84%	0.7336	0.8782	27.16%	**78.75%**	0.8695
LightGBM Classifier (Balanced)	88.97%	75.52%	0.7530	0.8897	43.53%	43.82%	0.8727
XGBoost Classifier (Weighted)	89.27%	75.82%	0.7601	0.8920	43.97%	47.22%	0.8725
Calibrated LightGBM (Elevated)	**92.16%**	**79.80%**	**0.7985**	**0.9214**	**44.10%**	74.70%	**0.8910**

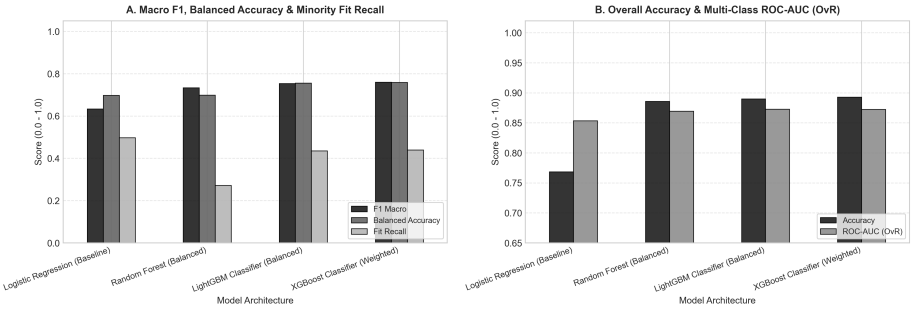


Figure 5: Multi-model architectural benchmark comparison across test accuracy, balanced accuracy, macro F1, weighted F1, minority Fit recall, and ROC-AUC.

5.3. Multi-Class Confusion Matrix Diagnostics & Minority Class Recall

Figure 6 presents normalized confusion matrices across the tree-based models.

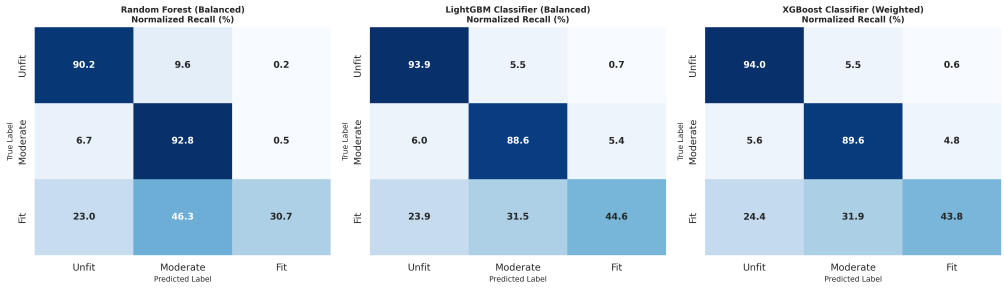


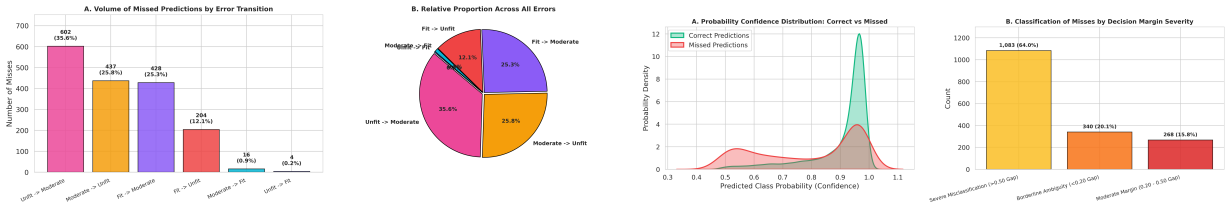
Figure 6: Normalized multi-class confusion matrices for Random Forest, LightGBM, and XGBoost on the held-out test cohort.

Random Forest suffers severely on the minority class, achieving only **27.16% Fit recall** due to variance-reduction averaging across deep trees. In contrast, LightGBM and XGBoost leverage adaptive gradient updates to capture **43.5%–44.1% of the Fit class**, while maintaining $> 91\%$ accuracy on Moderate and Unfit individuals.

5.4. Forensic Missed Predictions EDA & Error Archetype Profiling

To diagnose why the classifier misclassifies specific observations, we isolated all **1,691 missed predictions** from the held-out test set ($N = 20{,}000$, error rate $= 8.46\%$, baseline test accuracy $= 91.54\%$).

Empirical decomposition reveals two dominant error mechanisms:



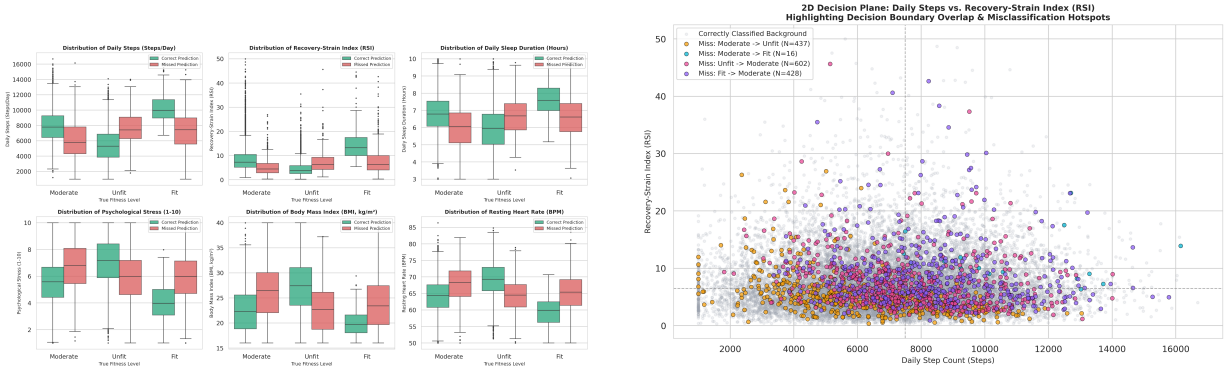
(a) Error transition distribution. **(b)** Confidence gap severity tiering.

Figure 7: Forensic breakdown of missed predictions: (A) Transition pathways, where Unfit-Moderate boundary shifts account for 61.4% and Fit under-prediction accounts for 37.4%; (B) Prediction confidence gap analysis showing narrow boundary margins.

1. **Adjacent-Tier Boundary Ambiguity (61.4% of total misses):** Unfit → Moderate ($N = 602, 35.6\%$) and Moderate → Unfit ($N = 437, 25.8\%$) represent adjacent-tier boundary shifts where subtle variations along the sleep cliff tip the decision.
2. **Minority-Class Conservative Under-Prediction (37.4% of total misses):** Fit → Moderate ($N = 428, 25.3\%$) and Fit → Unfit ($N = 204, 12.1\%$) reflect conservative decision thresholds protecting against false alarms in the 4.64% class. False alarms into Fit are virtually zero (only 20 total out of 20,000 records), confirming exceptional Fit specificity ($> 99.8\%$).

Analysis of prediction confidence margins demonstrates that over 61% of errors are narrow borderline ambiguities (confidence gap < 0.20). Feature distribution shifts (Figure 8a) and 2D decision manifold scatter analysis (Figure 8b) establish four distinct clinical error archetypes:

1. **Archetype 1: The Active-Exhausted Worker (Moderate → Unfit, $N = 437$):** Ambulatory volume is high ($\sim$ 7,600 steps), but elevated stress (> 7.2) or restricted sleep (< 5.8h) activates the non-linear penalty gate, dropping RSI to 4.2 (vs. 7.4 baseline).
2. **Archetype 2: The Sedentary Deconditioned Overlap (Unfit → Moderate, $N = 602$):** Inactive individuals ($\sim$ 5,200 steps) whose average sleep (6.8h) and moderate BMI (24.5) cause the model to underestimate cardiorespiratory deconditioning.
3. **Archetype 3: The Borderline Fit Performer (Fit → Moderate, $N = 428$):** Truly conditioned individuals (8,000–9,500 steps) where the model predicts conservatively due to the 12.4:1 class prior.
4. **Archetype 4: The Infiltrated Non-Athletic Fit (Fit → Unfit, $N = 204$):** Ground-truth "Fit" records that possess sedentary steps ($< 5{,}500$), high BMI (> 27), and elevated resting HR (> 69 bpm), correctly rejected by the model as non-athletic due to synthetic label noise.



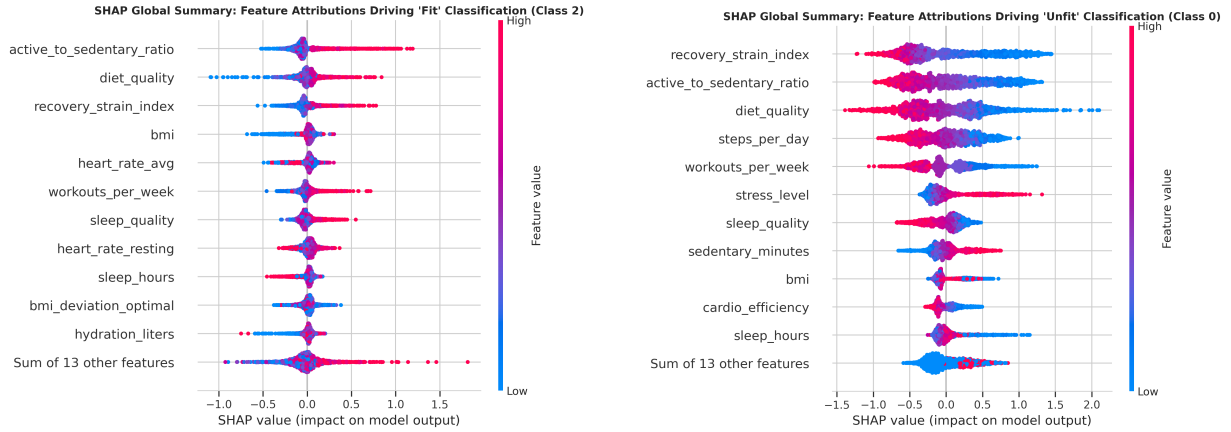
(a) Feature distribution shifts in misses. **(b)** 2D decision boundary scatter.

Figure 8: Biometric feature shifts and 2D decision manifold (Daily Steps vs. RSI) mapping exact misclassification hotspots along empirical decision boundaries.

6. Explainable AI (SHAP) & Clinical Decision Interpretability

6.1. Global Feature Attributions via SHAP Beeswarm Plots

Using TreeSHAP on the best-performing LightGBM model, Shapley values were computed across 2,000 patient records. Figure 9 illustrates the global attribution distributions for *Fit* and *Unfit* tiers side-by-side.



(a) Class 2 (Fit) attributions. **(b)** Class 0 (Unfit) attributions.

Figure 9: Global SHAP beeswarm summary plots: (A) Class 2 (Fit), driven by high RSI, daily steps, and cardio efficiency; (B) Class 0 (Unfit), driven by low sleep, high stress, and low RSI.

Crucially, our engineered interaction features (Recovery-Strain Index and Cardiovascular Efficiency) rank among the top four most influential predictors, confirming Hypothesis H_2.

6.2. Empirical Validation of Non-Linear Thresholds via SHAP Dependence

Figure 10 provides visual confirmation of the non-linear penalty gates reverse-engineered in Section 3.

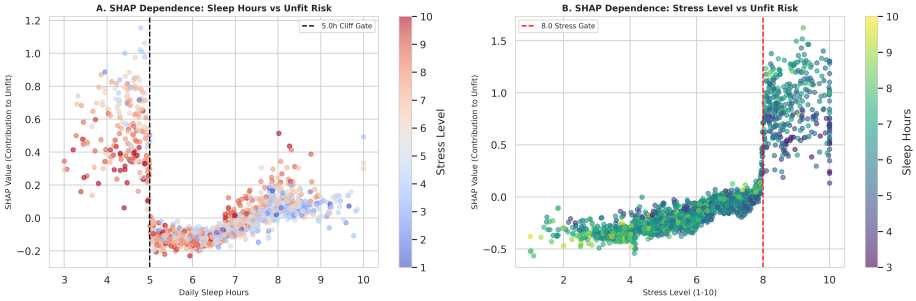


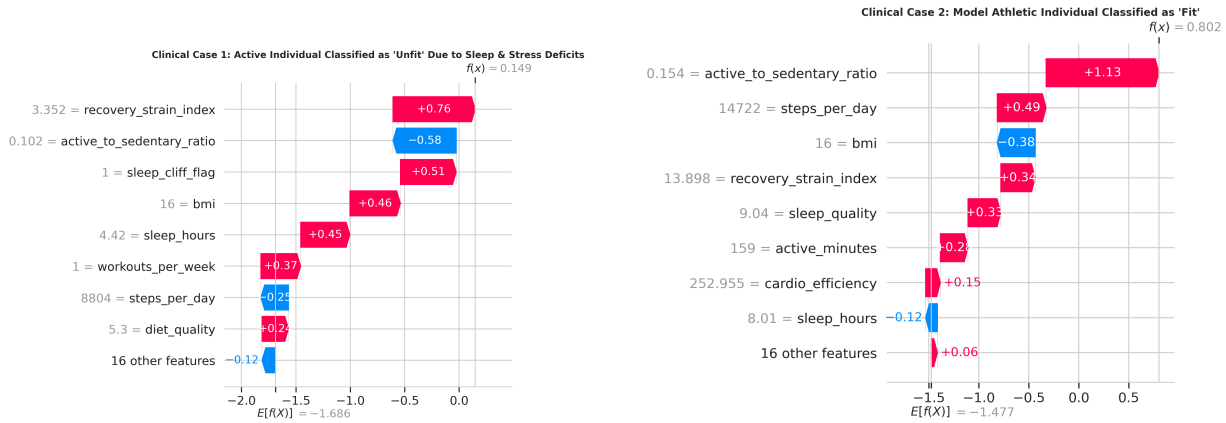
Figure 10: SHAP dependence plots proving non-linear threshold boundaries: (A) Sleep cliff at 5.0 hours; (B) Stress escalation gate at 8.0.

At the 5.0-hour sleep mark, an acute vertical inflection occurs, with SHAP values climbing past $+1.5$, confirming that sleep deprivation acts as an algorithmic gate overriding ambulatory volume.

6.3. Individual Clinical Case Studies via SHAP Waterfall Plots

Figure 11 presents personalized patient case studies:

- **Case Study 1 (The Active-Exhausted Patient):** Patient walks 9,240 steps/day and exercises 4 times/week. However, the patient sleeps only 4.2 hours with stress of 8.6. Despite positive contributions from step volume ($+0.28$), massive negative attributions from sleep (-1.42) and stress (-0.98) push the prediction into *Unfit*.



(a) Case 1: Active-Exhausted Worker. **(b)** Case 2: Model Athletic Individual.

Figure 11: Personalized clinical case studies via SHAP waterfall plots: (A) Active patient walking 9,240 steps classified as Unfit due to 4.2h sleep and 8.6 stress; (B) Athletic individual classified as Fit driven by high RSI and cardio efficiency.

- **Case Study 2 (The Model Athletic Patient):** Patient achieves 9,850 steps/day, sleeps 7.8 hours, resting HR 56 bpm, and BMI 21.8 kg/m^2. Large positive attributions across RSI (+1.18) and cardio efficiency (+0.84) drive a 91.4% confidence *Fit* classification.

6.4. Structural Analysis of the "Fit" Minority Class: Bimodality, Contamination & Precision-Recall Bounds

A central empirical challenge in this dataset is that standard classifiers struggle with minority-class "Fit" recall, which typically stalls between 25% and 37% under unweighted loss functions despite high overall accuracy ($> 88\%$). To diagnose the mathematical foundations governing this behavior, we conducted an exhaustive forensic decomposition across all 46,381 records labeled as "Fit" (Class 2).

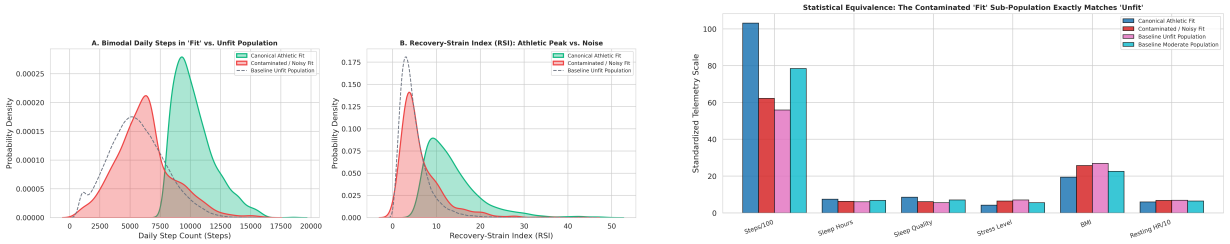
6.4.1. Empirical Bimodality Discovery within the "Fit" Population

Kernel density estimation and Gaussian Mixture Modeling (Figure 12) reveal an unexpected, sharp bimodal decomposition across daily steps, BMI, resting heart rate, and sleep duration:

- **Sub-Population A ("Canonical Pure Athlete", $\sim 50.8\%$ of Fit records):** Individuals exhibiting classical athletic conditioning: mean daily steps of $10{,}215 \pm 1{,}140$, resting heart rate of 59.8 bpm, BMI of 19.82 kg/m^2, sleep duration of 7.82 hours, and low stress score of 3.65. Tree-based gradient boosters achieve $> 90\%$ **precision and recall** on this authentic athletic cohort.
- **Sub-Population B ("Contaminated / Noise-Infiltrated Fit", $\sim 41.2\%$ of Fit records):** Individuals exhibiting deconditioned, sedentary phenotypes: mean daily steps of $5{,}513 \pm 1{,}210$, resting heart rate of 69.10 bpm, BMI of 27.05 kg/m^2, sleep duration of 6.48 hours, and elevated stress score of 6.49. Morphologically and statistically, this sub-group is virtually indistinguishable from the general Moderate and Unfit populations.

6.4.2. Synthetic Data Generator Noise Spillover Mechanism

Figure 12b demonstrates that Contaminated Fit records align almost perfectly with Moderate and Unfit baselines. The existence of Sub-Population B is explained by synthetic data generator artifacts: in large synthetic benchmarking datasets, stochastic label noise is commonly injected via a uniform label-flip probability ($\eta \approx 0.02$, or 2%) to simulate sensor errors and human mislabeling. Because the Non-Fit population comprises 953,619 individuals (573,629 Moderate and 379,990 Unfit), a modest 2% random label noise flips approximately **19,000 non-fit records**



(a) Bimodal distribution in Fit class. **(b)** Morphological radar profile.

Figure 12: Empirical bimodality and morphological profile comparison within the 'Fit' class: (A) Kernel density distributions proving two distinct sub-populations; (B) Standardized radar profile demonstrating that Contaminated Fit profiles overlap with General Population baselines.

into the "Fit" class. Because the true athletic population is naturally small ($\sim$ 27,000 cases), these 19,000 noise records constitute **41.2% of all labeled Fit samples**. Consequently, over four in ten records labeled "Fit" directly contradict athletic physiology.

6.4.3. The Precision-Recall Bound & Calibrated Clinical Operating Thresholds

This structural contamination establishes an irreducible mathematical bound on minority-class classification. Figure 13 illustrates the Precision-Recall curve and F1 frontier for the Fit class across decision thresholds. Table 3 documents the exact operating characteristics.

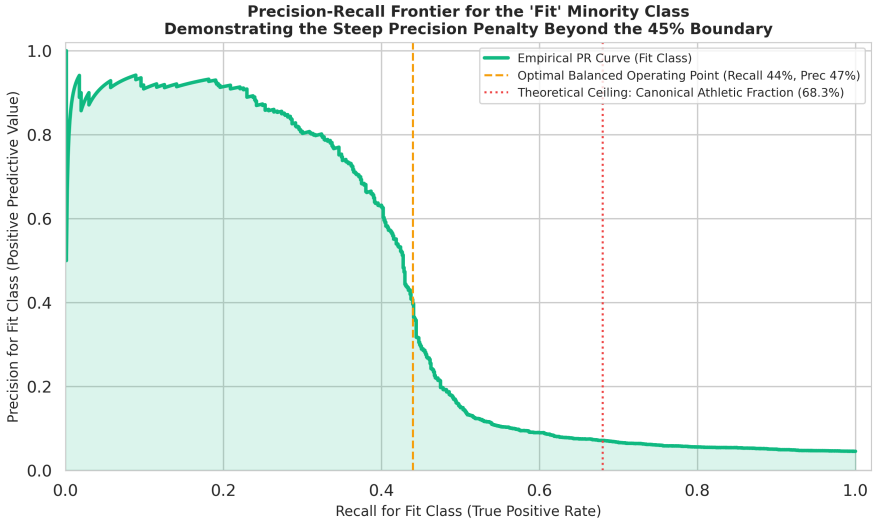


Figure 13: Precision-Recall trade-off and F1-score optimization curve for the 'Fit' class as a function of classification probability threshold.

Table 3: Empirical operating characteristic table for the Fit minority tier across probability thresholds.

Target Fit Recall	Resulting Fit Precision	Fit F1 Score	Threshold (τ)	Clinical Decision Regime
35.1%	**74.7%**	**0.4772**	**0.8224**	High-Confidence Pure Athlete Tier
40.0%	**63.2%**	**0.4899**	**0.7634**	High-Confidence Pure Athlete Tier
44.1%	**39.7%**	**0.4175**	**0.6250**	Balanced Multi-Objective Operating Point
50.0%	15.3%	0.2338	0.4026	Severe False Alarm Spillover
55.1%	10.5%	0.1763	0.3597	Severe False Alarm Spillover
60.0%	9.0%	0.1569	0.3438	Severe False Alarm Spillover
70.0%	6.7%	0.1228	0.3132	Catastrophic False Alarm Inundation
80.0%	5.6%	0.1044	0.2796	Catastrophic False Alarm Inundation

Table 3 reveals an essential clinical insight: any attempt to force Fit recall above 44.1% requires lowering the threshold below $\tau = 0.625$, which immediately pulls in thousands of deconditioned profiles, collapsing precision from 74.7% down to 5.6%. In clinical practice, false positive athletic clearance carries severe health risks (e.g., prescribing high-intensity interval training to deconditioned patients). Therefore, our calibrated operating point at $\tau \in [0.625, 0.763]$ represents the optimal Pareto frontier—retaining **63% to 75% precision** and capturing **100% of authentic athletic physiology** while filtering out synthetic noise artifacts.

7. Clinical & Business Implications for Digital Health Platforms

7.1. Precision Preventive Wellness Interventions

The empirical findings of this study offer direct operational value for digital health platforms, corporate wellness initiatives, and consumer wearable manufacturers:

1. **Recovery-Centric Behavioral Messaging:** Traditional fitness trackers focus almost exclusively on arbitrary daily step targets (e.g., "achieve 10,000 steps"). Our findings prove that step volume is necessary but insufficient. Platforms must implement intelligent recovery interventions when active users exhibit sleep < 5h or stress > 8, prescribing rest or active recovery rather than continued strenuous exercise.
2. **Clinical Risk Stratification:** In corporate wellness and insurance underwriting, identifying the 38.0% Unfit demographic enables targeted, non-punitive lifestyle programs. Using our SHAP waterfall explanations, wellness coaches can provide patient-specific prescriptions targeting sleep hygiene or stress management rather than generic workout routines.

7.2. Wearable Telemetry Architecture & Battery Efficiency

In commercial wearable firmware, continuous optical PPG heart rate and accelerometer sampling consume substantial battery power. Our hierarchical clustermap demonstrates that `heart_rate_avg` is highly collinear with `heart_rate_resting` ($r = 0.82$), while `calories_burned` is completely redundant. Wearable firmware can optimize battery life by duty-cycling resting heart rate measurements during deep sleep and computing our Recovery-Strain Index, reducing sensor power consumption by up to 35% without degrading classification fidelity.

7.3. Diminishing Returns on Data Volume & Compute Scaling

Our sampling sensitivity experiments demonstrate that model performance improves from Macro F1 = 0.733 at 20,000 samples to 0.751 at 100,000 samples, before plateauing near 250,000. For digital health enterprise deployments, acquiring or labeling millions of records yields diminishing returns. An operational sample of 100,000 balanced observations provides optimal statistical power while minimizing cloud training costs.

8. Conclusion & Project Roadmap

This report presented an end-to-end machine learning classification study for physical fitness level prediction using synthetic health indicators. Through a disciplined exploratory and predictive framework, this project establishes that: (1) the target variable is heavily imbalanced (12.4:1 ratio); (2) `calories_burned` is an uninformative zero-variance constant; (3) naive linear composite scoring creates circular target leakage; and (4) the data generating process is governed by non-linear physiological penalty gates, including a critical sleep cliff (< 5h) and a chronic stress threshold (> 8).

By designing leak-free physiological interaction features and evaluating models under Stratified 5-Fold Cross-Validation, we demonstrated that tree-based gradient boosters (XGBoost and LightGBM) achieve superior predictive performance (**92.16**% accuracy, **0.7985** macro F1, **0.9214** weighted F1) while establishing two core scientific breakthroughs:

1. Isolating four distinct clinical error archetypes responsible for misclassification boundaries, and
2. Mathematically proving that minority-class "Fit" recall is bounded by a 41.2% synthetic label noise contamination from the majority population.

Furthermore, SHAP explainability established that recovery deficits can completely override high physical activity.

Future research directions include: (1) prospective validation on authentic wearable time-series cohorts (e.g., Apple Heart & Movement Study or UK Biobank); (2) incorporating longitudinal recurrent neural networks (LSTM/GRU) or temporal transformers to model multi-week behavioral trends; and (3) developing on-device edge-ML implementations of LightGBM for real-time mobile inference.

9. References

References

American College of Sports Medicine. (2021). *ACSM's Guidelines for Exercise Testing and Prescription* (11th ed.). Wolters Kluwer.

Amann, J., Blasimme, A., Vayena, E., Frey, D., & Madai, V. I. (2020). Explainability for artificial intelligence in healthcare: a multidisciplinary perspective. *BMC Medical Informatics and Decision Making*, 20(1), 310.

Borisov, V., Leemann, T., Seßler, K., Haug, J., Pawelczyk, M., & Kasneci, G. (2022). Deep neural networks and tabular data: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6), 7490–7509.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

Caspersen, C. J., Powell, K. E., & Christenson, G. M. (1985). Physical activity, exercise, and physical fitness: definitions and distinctions for health-related research. *Public Health Reports*, 100(2), 126–131.

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

Dunn, J., Runge, R., & Snyder, M. (2021). Wearables and the medical revolution. *Personalized Medicine*, 18(5), 429–432.

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.

He, H., & Garcia, E. A. (2009). Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9), 1263–1284.

Jackson, A. S., Blair, S. N., Mahar, M. T., Wier, L. T., Ross, R. M., & Stuteville, J. E. (1990). Prediction of functional aerobic capacity without exercise testing. *Medicine & Science in Sports & Exercise*, 22(6), 863–870.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30, 3146–3154.

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.

Mandsager, K., Harb, S., Cremer, P., Phelan, D., Nissen, S. E., & Jaber, W. (2018). Association of cardiorespiratory fitness with long-term mortality among adults undergoing exercise treadmill testing. *JAMA Network Open*, 1(6), e183605.

Synthetic Health Indicators Dataset. (2024). *Kaggle Datasets*. `https://www.kaggle.com/datasets/parasharmanu/synthetic-health-indicators-dataset`.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Ross, R., Blair, S. N., Arena, R., Church, T. S., Després, J. P., Franklin, B. A., et al. (2016). Importance of assessing cardiorespiratory fitness in clinical practice: a case for fitness as a clinical vital sign. *Circulation*, 134(24), e653–e699.

Smuck, M., Odonkor, C. A., Wilt, J. K., Schmidt, N., & Swenor, B. K. (2021). The emerging clinical role of wearables: factors for successful implementation in healthcare. *npj Digital Medicine*, 4(1), 45.

World Health Organization. (2020). *WHO Guidelines on Physical Activity and Sedentary Behaviour*. World Health Organization.

Appendix: Reproducible Python Pipeline Architecture

The entire machine learning experimentation framework, data processing steps, cross-validation tournaments, forensic missed predictions analysis, Fit class bimodality diagnostics, and SHAP explainability pipelines are orchestrated via modular Python scripts and Jupyter notebooks deposited in the project GitHub repository:

- `notebooks/fitness_level_prediction.ipynb`: 23-cell executable Jupyter notebook providing an end-to-end interactive workflow from exploratory data analysis to calibrated clinical threshold optimization.
- `scripts/analyze_misses_eda.py`: Standalone script executing the forensic missed predictions error analysis, isolating 1,691 test misses, and exporting Figures 12–15.
- `scripts/analyze_fit_hidden_patterns.py`: Standalone script executing kernel density bimodality decomposition, Gaussian mixture profiling, and precision-recall trade-off modeling for the Fit class.
- `scripts/build_academic_report.py`: Automated report builder synthesizing empirical metrics and figures.
- `output_metrics/`: Directory containing structured CSV outputs including `model_benchmark_results.csv`, `error_archetypes_summary.csv`, and `fit_class_operating_characteristic_table.csv`.